Ratios and Rates

A **ratio** compares one number to another. A fraction is a kind of ratio that compares a part to a whole.

Example 1

Hide

Marcus has 15 coins: 5 quarters and 10 dimes. What is the ratio of quarters to coins?

There are 5 quarters out of 15 total coins. $\frac{5}{15} = \frac{1}{3}$

The ratio of quarters to total coins is $\frac{1}{3}$.

You can also write a ratio comparing the number of dimes to the number of quarters. Like all ratios, it can be written in three different forms.

Fraction Form	With a Colon	Using *to*
$\frac{10}{5}$	10:5	10 to 5

Ratios can be simplified to lowest terms, just like fractions. As with fractions, divide both numbers in the ratio by the same whole number.

Written in lowest terms: $\frac{10}{5} = \frac{2}{1}$ or 2:1 or 2 to 1.

The simplified ratio means that there are 2 dimes for every 1 quarter.

Ratios can be used to answer questions asking for a fraction of an amount.

Example 2

Hide

A local animal shelter rescues cats and dogs. The current ratio of cats to dogs is 12 to 15.
What fraction of the animals are cats?

Step 1
Add the number of cats and dogs to find the total number of animals.
12 + 15 = 27

Step 2
Find the number of cats in the ratio.
Cats are listed first in the ratio, so there are 12 cats.

Step 3
Write a fraction.
$\frac{12}{27} = \frac{4}{9}$

At the shelter, $\frac{4}{9}$ of the animals are cats.

Key Terms

- ratio
- unit rate
- cross products
- equivalent fractions

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Lesson 6: Ratios and Proportions

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